

Effects of Thiamine Supplementation on Exercise-induced Fatigue

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High-dose thiamine (vitamin B₁) supplementation (100 mg/day) may be helpful in preventing or accelerating recovery from exercise-induced fatigue. Sixteen volunteer male athletes volunteer, 8 with a blood thiamine level of 40 ng/ml or more (normal thiamine group) and 8 with levels below that level (low thiamine group) were selected as subjects. They exercised on a bicycle ergometer and the effects of thiamine supplementation were compared with placebo. Blood thiamine level markedly increased following supplementation of thiamine for 3 days before exercise. Exercise-induced changes in hemodynamic parameters and cardiopulmonary function indicated the onset of fatigue. Thiamine supplementation significantly suppressed the increase in blood glucose in the normal thiamine group and significantly decreased the number of complaints shortly after exercise in the subjective fatigue assessment of 30 items.

Keywords: thiamine, fursultiamine (TTFD), athletes, exercise-induced fatigue, blood glucose level, subjective fatigue symptoms

INTRODUCTION

Vitamin B₁ (thiamine) plays an important role as a coenzyme in energy metabolism and in the maintenance of nerve cell function (Itokawa, 1975; Von Mural, 1962). This suggests that thiamine might be effective not only in the prevention of or recovery from exercise-induced fatigue, but also in the improvement of mental and sensory fatigue, such as languor and malaise.

In the present study, healthy adult volunteers exercised on a bicycle ergometer and the effects of thiamine supplementation on the prevention of and recovery from exercise-induced fatigue were evaluated using a series of physiological measures and the subjective evaluation of fatigue.

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SUBJECTS and METHODS

A medical checkup was performed on 81 volunteer male athletes aged 19 years or more, and the blood thiamine level was determined. Sixty-one out of 81 male college-athlete volunteers had a thiamine level lower than 40 ng/ml and the mean blood thiamine level was 38.5 ± 8.8 ng/ml. Of these volunteers, 32 (40%) were shown to have a blood thiamine level of between 41 and 68 ng/ml and 49 (60%) to have a level less than 40 ng/ml. No individual had a blood thiamine level below 20 ng/ml. Eight individuals from both the normal (>40 ng/ml) and low (≤ 40 ng/ml) groups were selected as subjects (Table 1). All of the subjects were well informed of the nature and contents of this study by the investigator and gave their consent to participate in this study.

Table 1. Mean ($\pm$ SD) age, body weight and height and range of blood thiamine levels for normal and low thiamine groups.

	Normal Thiamine Group	Low Thiamine Group
No.	8	8
Age (yr.)	23 ± 2	21 ± 1
Height (cm)	172 ± 4	173 ± 6
Body weight (kg)	65 ± 4	64 ± 5
Blood thiamine (ng/ml)	56.8 ± 15.0	30.9 ± 7.7

Test drug: The thiamine preparation contained 100 mg of thiamine as fursultiamine (Thiamine Tetrahydrofurfuryl Disulfide, TTFD), thiamin derivative, as well as vitamins B₂ (10 mg), B₆ (10 mg), and B₁₂ (20 μ g), in 2 tablets. Placebo tablets were identical in appearance to the active tablets and contained lactose as the major component.

Experimental schedule: A comparative study of the thiamine preparation versus placebo was performed in a cross-over manner (Table 2). The subjects took 2 tablets of the vitamin preparation or placebo with water after breakfast for 3 days prior to the day of exercise. They took 2 more tablets after supper on the day of exercise to facilitate the recovery from fatigue. Half of each group took the active tablets first, and the others took placebo tablets first. A washout period of 3-9 days was allowed between the completion of the first test session and the start of the second supplementation.

A medical interview to confirm general health and to draw blood samples was performed before thiamine supplementation and bicycle ergometer exercise. Meals and physical activity during the supplementation period were recorded.

Table 2. Thirty items were assessed for subjective fatigue symptoms

A. General Languishness	B. Spiritual Concentration	C. Physical Malaise (Local)
1. Heavy headedness	11. Difficulty in thinking	21. Headache
2. General dullness	12. No mood for chat	22. Stiff shoulder
3. Dullness of legs	13. Irritation	23. Lumbago
4. Tendency to yawn	14. Distraction	24. Choking
5. Absent-mindedness	15. Lack of zeal	25. Abnormal feeling in mouth
6. Sleepiness	16. Lapse of memory	26. Harsh voice
7. Fatigue of eyes	17. No confidence	27. Dizziness
8. Awkward behavior	18. Uneasiness	28. Twitching in eyes and muscles
9. Unsteady walk	19. Sloppiness	29. Trembling of hands and legs
10. Weakness	20. Lack of patience	30. Nausea

Assessment of thiamine effect: The subjects exercised on a bicycle ergometer for 30 min (60% of VO₂ Max.) at 1 h after breakfast of the day of testing. Evaluation items and hemodynamic parameters were measured before, immediately after (0 min), and at 30, 60, and 120 min after the exercise. In addition, subjective symptoms were recorded the next morning.

Subjective fatigue was evaluated as generalized fatigue on a scale of 5 and using a questionnaire of 30 items prepared by the Industrial Fatigue Study Group of the Japanese Society for Industrial Hygiene (Sangyo Hiro Kenkyukai) of Japan Society for Occupational Health (Table 2). In the latter evaluation, the subjects marked the items they felt, from the list of 30 items. A larger number denotes a more fatigued condition.

Statistical analysis: Category variables were analyzed by the Mann-Whitney U-test, and continuous variables by the t-test for cross-over design.

RESULTS

The results of hemodynamic parameters and the evaluation items are summarized as follows:

Biochemical Parameters and Cardiopulmonary Function

The blood thiamine levels markedly increased after supplementation of the thiamine preparation, indicating good absorption (Figure 1).

The levels of blood catecholamines (adrenaline, noradrenaline and dopamine) increased immediately after exercise, but returned to the pre-exercise levels after 1 h. Thiamine supplementation had no noticeable effect on the plasma levels of catecholamines, which are known to rise immediately after exercise (Fujii *et al.*, 1992; Miyagawa *et al.*, 1991). The blood glucose level, which is related to energy metabolism, increased during the first 30

min after exercise. This rise was significantly suppressed with thiamine supplementation in the normal thiamine group. Heart rate increased transiently after exercise but returned to the pre-exercise levels after 1 h. Oxygen consumption also rose after exercise and the levels after 2 h was still higher than the pre-exercise levels, though it declined steadily. No effect of thiamine supplementation was observed (Figure 2).

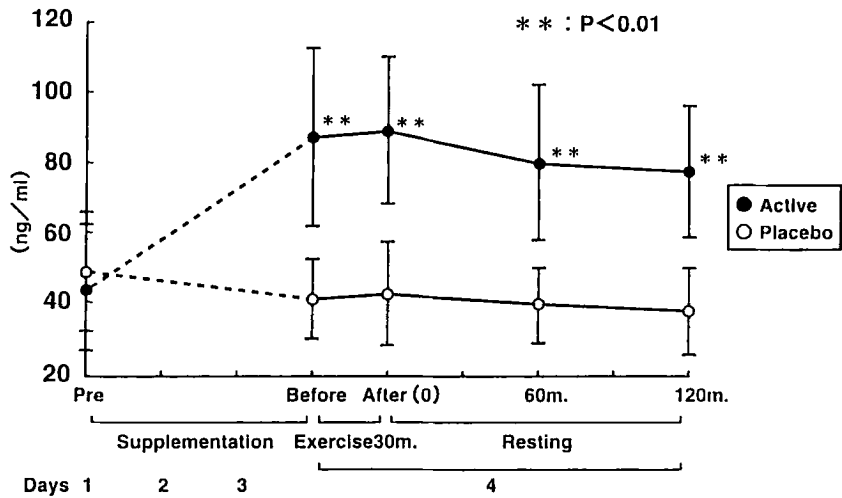


Figure 1. Blood thiamine in thiamine supplemented versus placebo-treated athletes.

Blood lactate levels increased following exercise in all subjects. The serum free fatty acid level decreased immediately after exercise and returned to the pre-exercise level after 2 h. Blood pressure increased transiently after exercise but returned to the pre-exercise levels after 1 h. Respiratory quotient was not affected by exercise. No effect of thiamine supplementation on these parameters was observed.

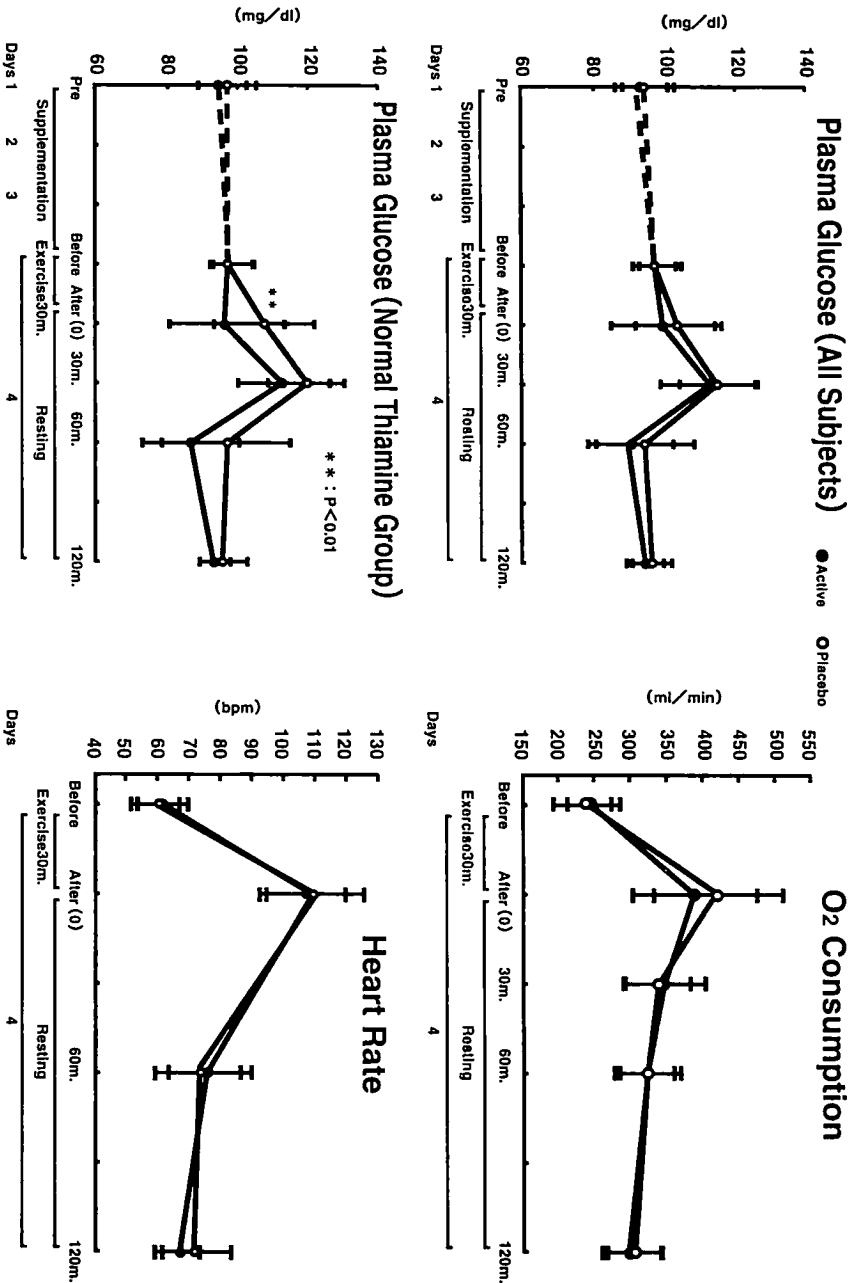


Figure 2. Physiological parameters in thiamine supplemented (●) versus placebo-treated (○) athletes.

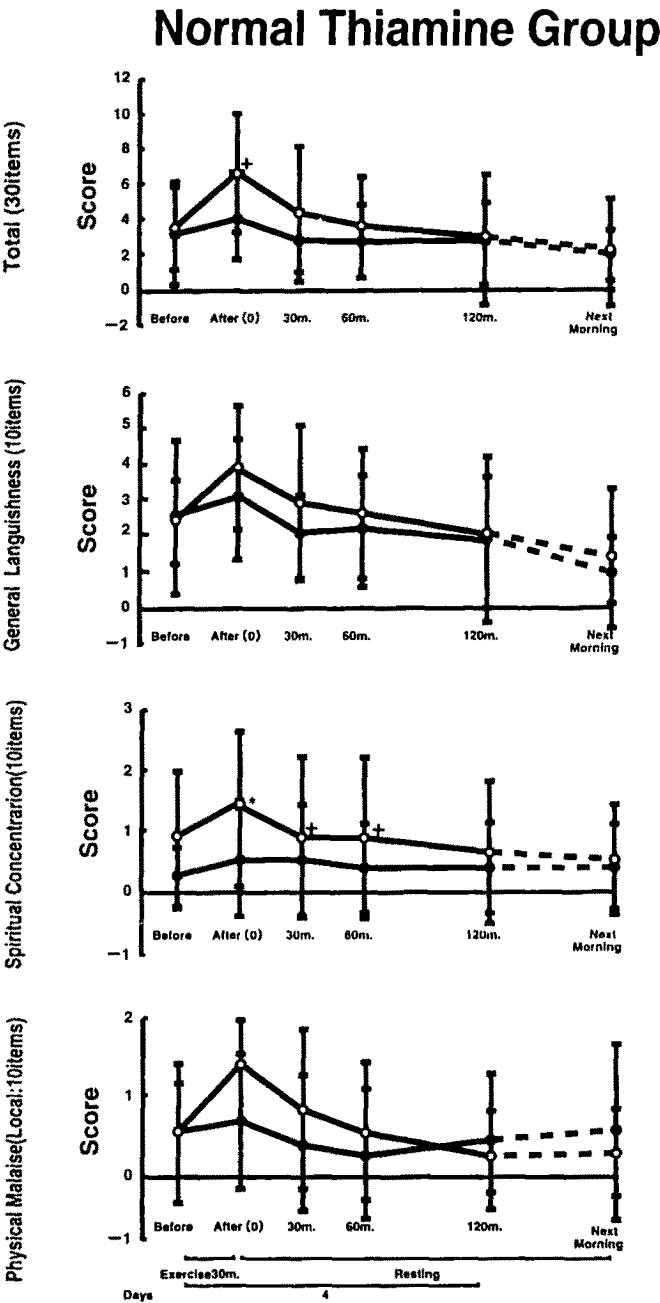


Figure 3a: Normal thiamine group. Subjective fatigue symptoms (30 items) in thiamine-supplemented (●) versus placebo-treated (○) athletes. * $p < 0.05$; + $p < 0.01$.

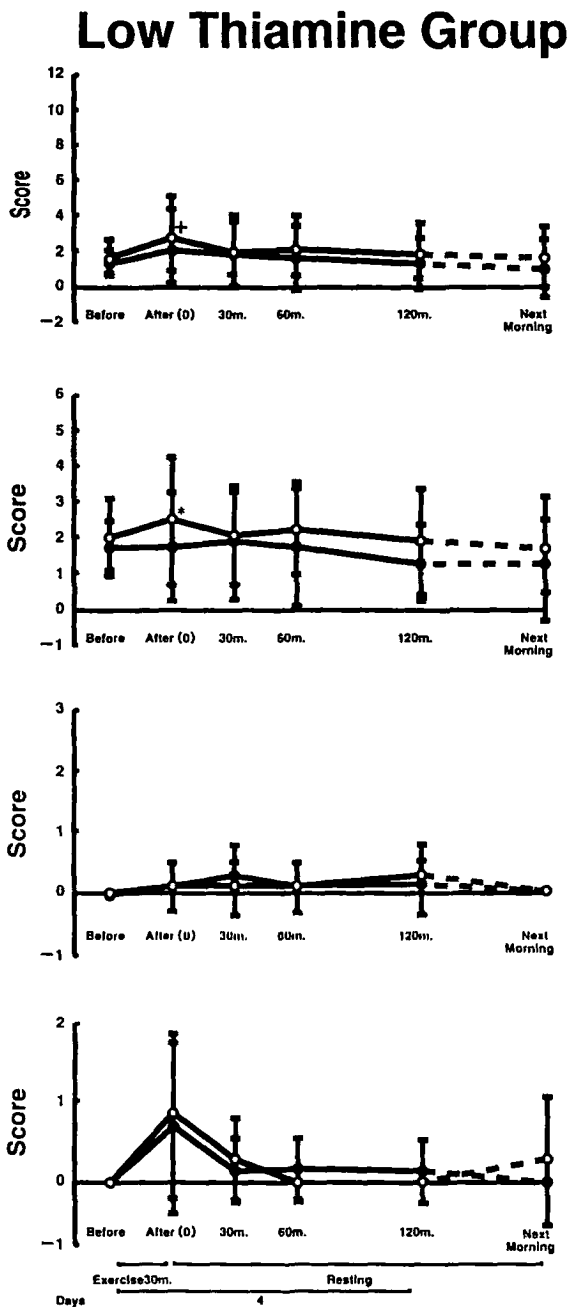


Figure 3b: Low thiamine group. Subjective fatigue symptoms (30 items) in thiamine-supplemented (●) versus placebo-treated (○) athletes. * $p < 0.05$; + $p < 0.01$.

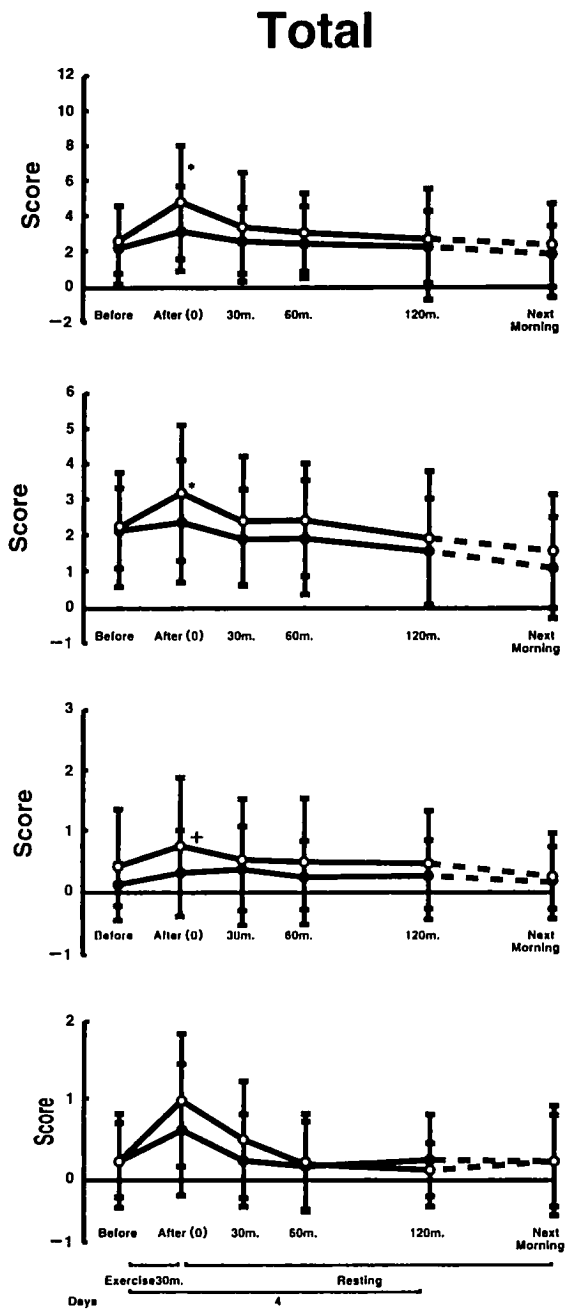


Figure 3c: Normal plus low thiamine group (Total). Subjective fatigue symptoms (30 items) in thiamine-supplemented (●) versus placebo-treated (○) athletes. * $p<0.05$; + $p<0.01$.

Subjective Findings

Subjective symptom evaluation using 30 items indicated significant improvement immediately after exercise when thiamine supplementation was given ($p<0.05$) (Figure 3). Statistical analyses revealed separately, no significant differences were noted in either normal thiamine group and low thiamine group, but subjective fatigue was significantly alleviated in both groups with thiamine supplementation ($p<0.01$).

In the ten items (A) in Table 2, which indicate general languishness, the effect of thiamine supplementation was significantly better in the low thiamine group and in the total of 2 groups (Figure 3).

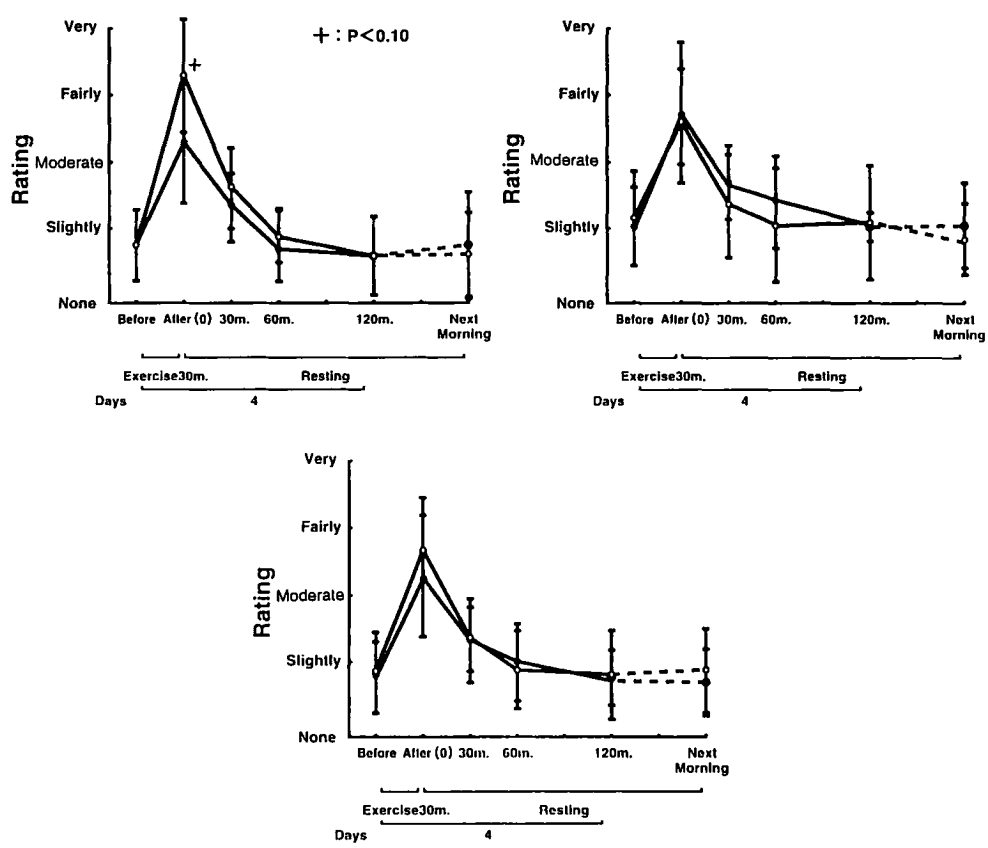


Figure 4. Overall fatigue rating in thiamine supplemented (o) versus placebo-treated (□) athletes.

In the ten items (B) in Table 2, which indicate spiritual concentration, thiamine supplementation significantly maintained spiritual concentration in the normal thiamine group. Spiritual concentration tended to be maintained with thiamine supplementation (Figure 3). The effect of thiamine supplementation on physical malaise was not significant (Figure 3). As for the overall physiological fatigue rating, fatigue rate tended to be less with thiamine supplementation in the normal thiamine group, but its effect in the low thiamine group and in the total of 2 groups was not significant (Figure 4).

DISCUSSION

Many aspects of fatigue have been studied over the years, but adequate parameters for objective evaluation of fatigue have not yet been established. In the present study, healthy adult male athletes exercised to fatigue, and the effect of thiamine supplementation on fatigue was then evaluated.

The results of a medical checkup performed before selection of subjects revealed that 61% (49/81) of those surveyed had a blood thiamine level less than the lower limit of normal (40 ng/ml), indicating a marginal or latent deficit. It is unknown whether or not the levels among athletes are different from those found in the general population. It is reported that oarsmen who consumed energy of 4000 kcal or more a day took a larger amount of thiamine than the recommended requirement, but 11 of 14 oarsmen were judged to have a marginal deficiency (Kobayashi, 1990). This suggests that the thiamine requirement of athletes may be larger than that recommended for non-athletes.

Fursultiamine (TTFD), a thiamine derivative, has been proven to be absorbed efficiently from the intestine, to have a large tissue distribution, and to be readily converted into thiamine. Therefore, TTFD is expected to have a greater physiological effect than conventional thiamine because of improved absorption (Itokawa *et al.*, 1992). TTFD has been widely used not only in the treatment of thiamine deficiency and various diseases with possible energy metabolism disorders but also to prevent or accelerate the recovery from exercise-induced fatigue. Studies have shown that TTFD, which was supplemented to physical workers in a high-temperature environment (Nakamura, 1971) and to athletes in a training camp for 10 days to 2 weeks (Nishiyama *et al.*, 1972), was effective in the recovery from chronic physical fatigue. When conventional thiamine preparations were supplemented (10 mg/day) to elderly people with marginal thiamine deficiency for 6 weeks, various benefits such as increased appetite, improved general well-being, and decreased fatigue were observed (Smidt, 1991). As shown in these reports, long-term supplementation of thiamine preparations has been demonstrated to be effective in accelerating the recovery from fatigue, but the effects of short-term thiamine supplementation (3 days) on recovery from acute fatigue due to single exercise have not yet been studied. Although the effects on recovery from fatigue should be evaluated on substances administered after the onset of fatigue, it was suggested in one report that thiamine must be stored in the muscles before exercise (Tsuji, 1991). Accordingly, in the present study, TTFD was supplemented for 3 days before exercise to evaluate its effect.

TTFD was also supplemented in the evening on the day of exercise and its effect on the recovery from fatigue was evaluated the next morning.

In the present study, the number of complaints regarding subjective fatigue were significantly decreased immediately after exercise by thiamine supplementation. In the 10 items relating to general languishness in Table 3, thiamine supplementation significantly alleviated malaise in the total of 2 groups and the low thiamine group. This finding substantiated our hypothesis that there might be a correlation between blood thiamine levels and fatigue, and it may support the needs of thiamine supplementation to people suspected of having insufficient thiamine levels, e.g. those who are unbalanced diet eaters, liable to fatigue, heavy alcohol drinkers, hard workers, competitive athletes, etc. In the 10 items relating to spiritual concentration, thiamine supplementation significantly suppressed the exercise-induced lowering of spiritual concentration in the normal thiamine group. The reason for this is unknown, though we might speculate that thiamine played a role other than just a supplementary one. But all in all, there was no clear difference in the effect of thiamine between the low thiamine and normal thiamine groups. This may be due to the fact that the thiamine in the low thiamine group is only marginally deficient, and that there was no marked difference in the blood thiamine level between the low thiamine (mean 30.9 ± 7.7 ng/ml) and normal thiamine (mean 56.8 ± 15.0 ng/ml) groups. A larger number of volunteers may be needed to study this subject further. Thiamine supplementation had no effect on subjective fatigue symptoms measured the morning after the exercise, but this may have been because the subjects were young athletes who could quickly recover from exercise-induced fatigue.

Thiamine supplementation influenced neither the increase in the plasma lactate level nor the decrease in the serum free fatty acid level caused by muscular exercise. Respiratory quotient, which is defined as the ratio of exhaled carbon dioxide gas to inhaled oxygen gas, was not affected by exercise. This phenomenon might be explained by the fact that the subjects were athletes and well trained. Thiamine supplementation did suppress the increase in the blood glucose level in the normal thiamine group. This finding is difficult to interpret, but thiamine, as a coenzyme, may enhance the energy metabolism of glucose in some way (Takahashi, 1981). Namely, ingestion of a sufficient amount of thiamine might increase pyruvate dehydrogenase activity in the glycolytic pathway and α -ketoglutarate dehydrogenase activity in the TCA cycle. Hence, thiamine may promote energy production by utilizing glucose efficiently as an energy source. It is unknown whether this possible effect on glucose is related to the suppressive effect of thiamine supplementation on the lowering of spiritual concentration seen in the normal thiamine group.

CONCLUSION

Sixteen young male athletes were divided into a low thiamine group (<40 ng/ml) and a normal thiamine group (≥ 40 ng/ml), and the effects of thiamine supplementation on exercise-induced fatigue were compared with placebo.

- 1) The blood thiamine level markedly increased following supplementation of thiamine for 3 days before exercise.
- 2) Exercise-induced changes in hemodynamic parameters and cardiopulmonary function indicated the onset of fatigue. Thiamine supplementation before exercise significantly suppressed the increase in blood glucose level in the normal thiamine group but had no effect on the other hemodynamic parameters or on cardiopulmonary function.
- 3) Regarding subjective fatigue assessed by 30 items, the number of complaints shortly after exercise significantly decreased following thiamine supplementation, indicating that thiamine either prevents or promotes recovery from fatigue.
- 4) In ten items indicating general languishness, thiamine supplementation alleviated fatigue complaints shortly after exercise all subjects as a whole and in the low thiamine group. In the ten items indicating spiritual concentration, thiamine supplementation significantly alleviated the lowering of spiritual concentration seen shortly after exercise in tall subjects as a whole and in the normal thiamine group. The mechanism by which thiamine produces these differences requires further study.

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